

Speech in NLP: Text-to-Speech (TTS)

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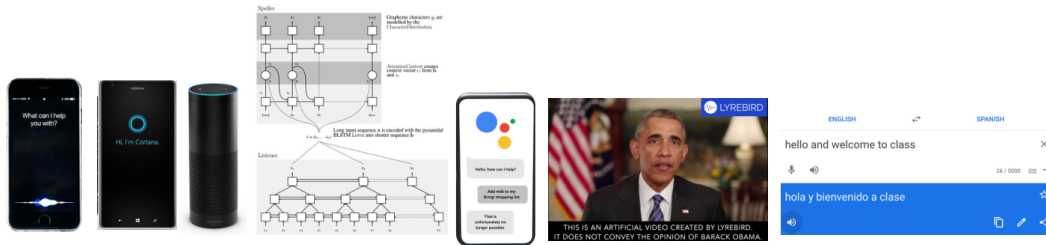
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*“Words mean more than what is set down on paper.
It takes the human voice to infuse them with shades of deeper meaning.”
— Maya Angelou, I Know Why the Caged Bird Sings*

Exciting Recent Developments have Disrupted the Field



2011:
Apple Siri

2014:
Microsoft Cortana
Amazon Alexa
Alexa Prize

2015:
End-to-end neural
becomes SORA

2016:
Google
assistant

2017:
Neural TTS voice cloning

2020:
Realtime speech-speech
translation

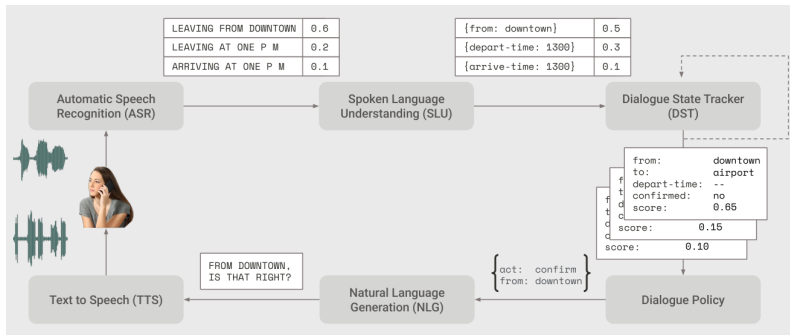


Figure 2: Architecture of dialogue-state system for task-oriented dialogue (Williams et al., 2016)

What is Text-to-Speech (TTS)?

- ▶ **What:** Produce speech from a text input (**Input:** text, optionally with speaker, language, style, or acoustic prompt; **Output:** waveform or codec tokens perceived as speech).
- ▶ **Applications:**
 - Accessibility (screen readers, assistive tech)
 - Personal Assistants (Apple Siri, Microsoft Cortana, Google Assistant)
 - Navigation, games, announcements/voice-overs, audiobooks, dubbing, AR/VR
 - Synthetic data for ASR

What is Text-to-Speech (TTS)?

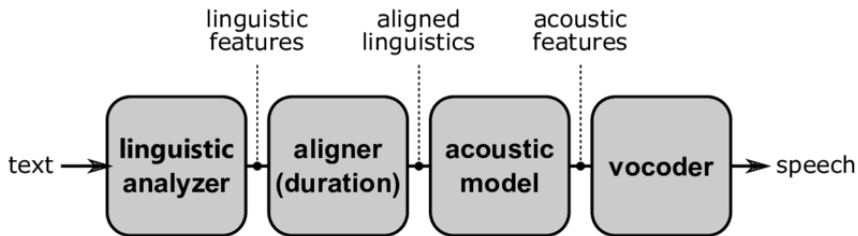
- ▶ **Core difficulty:** Many-to-many mapping: the same text admits many natural realizations (prosody, rate, emphasis).
- ▶ **New variation:** Voice cloning: TTS systems that mimic particular speakers with minimal training data.

- ▶ **Prosody** covers everything the words alone do not fix: intonation (the pitch contour, F_0), duration and rhythm, and energy (stress).
- ▶ Stress changes meaning: “**I** never said she stole my money” accuses differently than “I never said she stole **my** money”. Seven words, seven readings, one spelling.
- ▶ Intonation signals sentence type: statements tend to fall in pitch, yes/no questions tend to rise.
- ▶ TTS systems model prosody implicitly (attention, latent variables) or explicitly (FastSpeech 2 predicts duration, pitch, and energy per phoneme).

- ▶ Collect lots of speech (5–50 hours) from one speaker, transcribed carefully down to syllables and phones.
- ▶ Rapid recent progress in neural approaches.
- ▶ Modern systems are DNN-based, understandable, but not yet fully emotive.
- ▶ No longer require extensive data from a single speaker to make a TTS voice for them.

- ▶ **Codec:** A coder/decoder system that encodes analog speech signals into a digitized, discrete compressed representation for compression.
- ▶ The discrete symbols a codec produces as its compressed form can serve as **discrete codes** for language modeling.
- ▶ A codec typically consists of:
 - An **encoder** (e.g., convnets) that downsamples speech into an embedding,
 - A **quantizer** that converts the embedding into a series of discrete tokens,
 - A **decoder** (e.g., convnets) that upsamples the tokens/embedding back into a (lossy) reconstructed waveform.

1. **Text analysis:** normalization, tokenization, G2P
2. **Acoustic model:** mel, latents, or codec codes over time
3. **Vocoder / decoder:** waveform or neural codec inversion



- ▶ **Pre-2016:** concatenative unit selection and statistical parametric (HMM/GMM) synthesis.
- ▶ **2016–2021:** neural end-to-end models (Tacotron 2, WaveNet vocoders, FastSpeech, Glow-TTS, VITS).
- ▶ **2022–present:** diffusion, flow matching, and **foundation audio** models (VALL-E, Voicebox) with zero-shot cloning and in-context control.
- ▶ Each era targets bottlenecks of the previous one: flexibility, fidelity, latency, then controllability at scale.

Three eras of text-to-speech

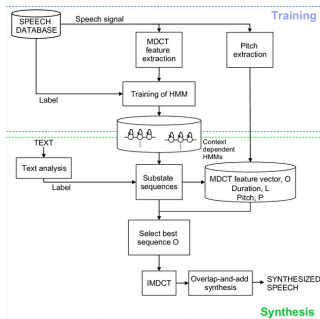
Era	Technology	Strength	Bottleneck
Concatenative	Unit / diphone selection	Local timbre	Huge DB; brittle joins
Statistical (SPSS)	HMM / GMM vocoder params	Small footprint; adaptation	Muffled, averaged sound
Neural (E2E)	Tacotron 2 + neural vocoder	High fidelity; simpler stack	Slow AR inference
Foundation	VALL-E, Voicebox	Zero-shot; in-context learning	Data scale; misuse risk

- ▶ Raw text is full of **non-standard words**: numbers, dates, times, currency, abbreviations, URLs.
- ▶ The same string reads differently by context:
 - “Dr. Smith lives on King Dr.” (doctor vs. drive)
 - “1995” as a year (nineteen ninety-five) or a quantity (one thousand nine hundred ninety-five)
 - “3/4” as a fraction (three quarters) or a date (March fourth)
- ▶ Classical solution: rules and lexicons per category; a large share of production TTS bugs start here.
- ▶ Modern systems use neural taggers or fold normalization into the model, but deployed stacks keep rule layers for safety.

- ▶ English spelling underdetermines pronunciation: *though, tough, through, and thought* share letters, not sounds.
- ▶ **Homographs** need context to resolve: “I *read* it yesterday” vs. “I *read* every day”; *bass* the fish vs. *bass* the instrument.
- ▶ Standard pipeline: pronunciation lexicon lookup (CMUdict, about 134k ARPAbet entries) plus a neural G2P model for out-of-vocabulary words and names.
- ▶ Phoneme input makes synthesis more stable; character-based models (Tacotron reads characters) learn G2P implicitly and can mispronounce rare words.

- ▶ Early systems used **unit selection**: recording lots of speech from a single speaker and splitting it into small parts (like diphones or syllables).
- ▶ During synthesis, an algorithm picks which units to use by balancing:
 - **Target cost** (how well each piece matches the desired sound)
 - **Join cost** (how smooth the pieces sound when joined together)
- ▶ This gives good local naturalness, but is not very flexible and can sound glitchy if needed transitions are missing from the database.

- ▶ Statistical Parametric Speech Synthesis (SPSS) mainly used HMMs to model speech features instead of storing raw audio.
- ▶ SPSS models capture the spectral envelope, fundamental frequency (F_0), and duration of speech segments.
- ▶ Speaker adaptation is possible by updating a pre-trained model with a small amount of new speaker data, using MAP or linear regression techniques.
- ▶ Trajectory HMMs were introduced to make transitions between features smoother and output speech more natural.

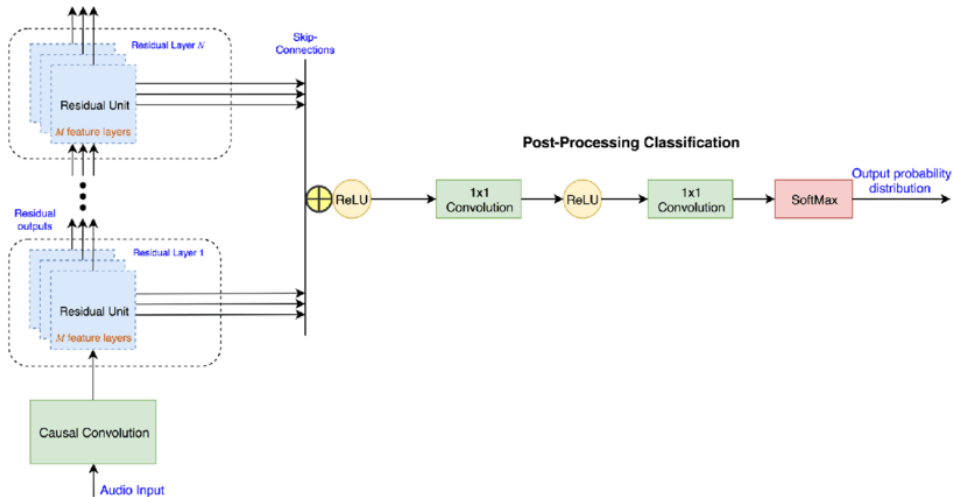


- ▶ Deep generative model for raw audio
- ▶ Autoregressive: predicts next sample given previous samples
- ▶ High-quality, natural-sounding speech
- ▶ Equation:

$$p(x) = \prod_{t=1}^T p(x_t | x_1, \dots, x_{t-1})$$

- ▶ Computationally expensive for real-time use

2016–2021 Era: The Neural Revolution (WaveNet) (cont.)



Source: van den Oord et al., WaveNet, after DeepMind.

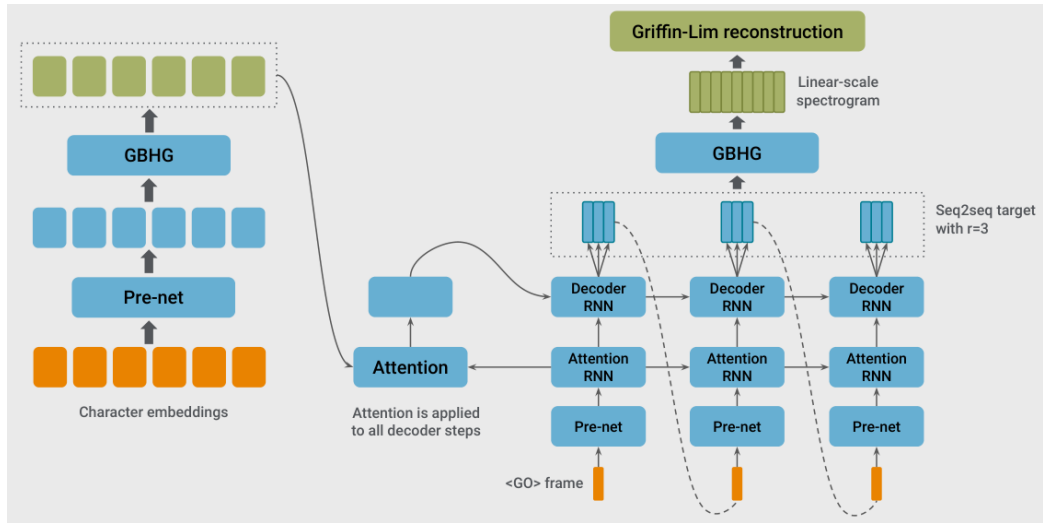
- ▶ Mel-spectrograms show how frequencies change over time, but must be turned back into sound waveforms.
- ▶ The Griffin-Lim algorithm was an early method for this, but the quality was limited.
- ▶ Neural vocoders like WaveNet can reach human-level speech quality.
- ▶ WaveNet generates raw audio samples one by one, at up to 24,000 times per second.
- ▶ It uses dilated causal convolutions, so it can “see” a lot of context efficiently.
- ▶ WaveNet models the conditional probability of each sample:

$$p(\mathbf{x} \mid \mathbf{h}) = \prod_{t=1}^T p(x_t \mid x_1, \dots, x_{t-1}, \mathbf{h})$$

where $\mathbf{h}$ is the mel-spectrogram.

- ▶ This approach gives very natural speech, but it is slow to run for real-time use.

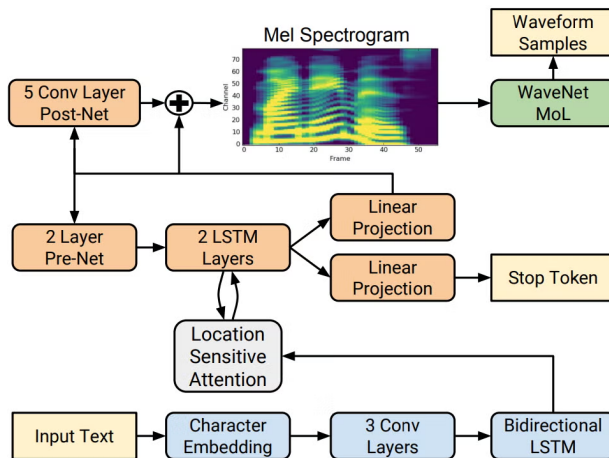
2016–2021 Era: End-to-end Neural TTS (Tacotron)



Source: Tacotron (Wang et al, 2017); Model Architecture. The model takes characters as input and outputs the corresponding

- ▶ End-to-end TTS system
- ▶ Pipeline:
 - Text $\rightarrow$ Mel-spectrogram (sequence-to-sequence model)
 - Mel-spectrogram $\rightarrow$ WaveNet/GAN vocoder (audio synthesis)
- ▶ Improved prosody and pronunciation
- ▶ Flexible for different voices and languages

Tacotron 2 (cont.)



Source: Tacotron 2, after Shen et al., 2018..

- ▶ Tacotron 2 + WaveNet-class vocoders raised fidelity but blocked low-latency, on-device synthesis.
- ▶ Motivated **non-autoregressive (NAR)** acoustic models and fast vocoders (FastSpeech, HiFi-GAN).

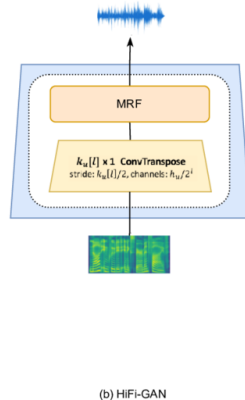
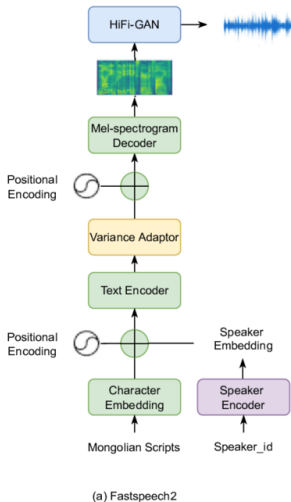


- ▶ Non-autoregressive models: faster inference
- ▶ FastSpeech:
 - Parallel generation of spectrograms
 - Duration prediction for alignment
- ▶ HiFi-GAN:
 - GAN-based vocoder
 - High-fidelity, real-time synthesis

- ▶ Glow-TTS:
 - Flow-based generative model
 - Efficient and expressive speech synthesis

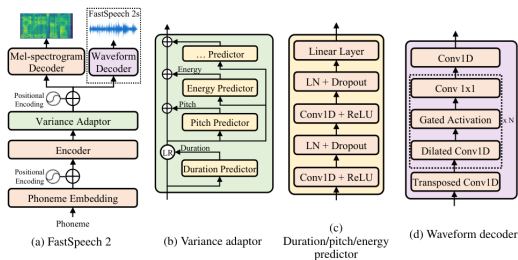
- ▶ Diffusion models:
 - Iterative refinement of audio
 - State-of-the-art naturalness

Beyond Autoregression: FastSpeech, HiFi-GAN, Glow-TTS, Diffusion Models (cont.)



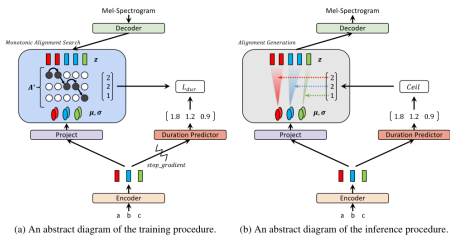
FastSpeech 2: parallel mel generation

- ▶ FastSpeech 2 uses a Feed-Forward Transformer (FFT) with a Length Regulator to match phoneme and spectrogram lengths, replacing soft attention.
- ▶ The Variance Adaptor predicts duration, pitch, and energy for more control over prosody.
- ▶ The Duration Predictor solves the "one-to-many" mapping, deciding exactly how many frames each phoneme lasts, which makes the system more stable.
- ▶ Pitch Predictor models the fundamental frequency (F_0); Energy Predictor models the loudness (from the spectrogram).



Source: Ren et al., "FastSpeech 2: Fast and High-Quality End-to-End Text to Speech," Fig. 1(a); [arXiv:2006.04558](https://arxiv.org/abs/2006.04558) (cropped).

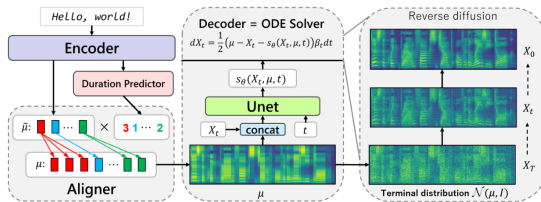
- ▶ Glow-TTS uses Normalizing Flows to turn a simple distribution into a mel-spectrogram distribution.
- ▶ Monotonic Alignment Search (MAS) finds the single best monotonic path between text and speech using dynamic programming.
- ▶ Unlike soft attention, MAS avoids alignment breaks, so the model works well on long utterances.



Source: Kim et al., "Glow-TTS: A Generative Flow for Text-to-Speech via Monotonic Alignment Search," Fig. 1;

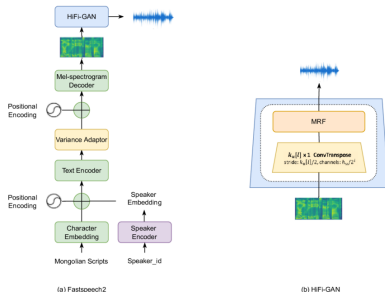
[arXiv:2005.11129](https://arxiv.org/abs/2005.11129).

- ▶ Grad-TTS uses score-based diffusion to turn random noise into structured mel-spectrograms.
- ▶ The reverse diffusion process is controlled by stochastic differential equations (SDEs).
- ▶ Users can control the trade-off between quality and speed by choosing the number of reverse steps.
- ▶ Grad-TTS can sound good with as few as 10 steps, and it is faster and more robust than Tacotron 2.



Source: Popov et al., "Grad-TTS: A Diffusion Probabilistic Model for Text-to-Speech," Fig. 2; [arXiv:2105.06337](https://arxiv.org/abs/2105.06337).

- ▶ HiFi-GAN is a GAN-based vocoder that can generate high-quality audio from mel-spectrograms.
- ▶ Parallel front ends like FastSpeech can generate mel-spectrograms in parallel, which is faster than sequential generation.
- ▶ The combination of FastSpeech and HiFi-GAN is a popular and effective way to synthesize speech.

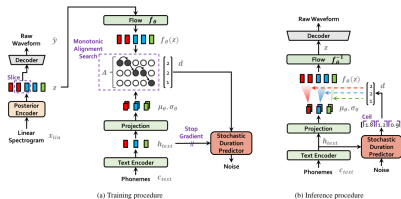


Source: Kong et al., "Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis," Fig. 1;

[arXiv:2010.05646](https://arxiv.org/abs/2010.05646).

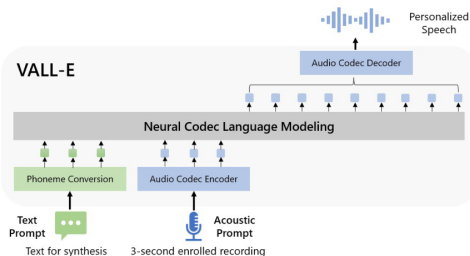
VITS: Variational Inference with Adversarial Learning for End-to-End Text-to-Speech

- ▶ Connect two modules of TTS systems through latent variables to enable efficient end-to-end learning.
 - generates more natural sounding audio than current two-stage models
- ▶ At training: a posterior encoder reads real melts; monotonic alignment search (MAS) links text to speech; a discriminator improves waveform quality.
- ▶ At inference: text $\rightarrow$ text encoder + duration predictor $\rightarrow$ latent features $\rightarrow$ waveform decoder $\rightarrow$ audio (no separate vocoder).



Source: Kim, Kong, and Son, "Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech" (VITS), Fig. 1; [arXiv:2106.06103](https://arxiv.org/abs/2106.06103).

- ▶ VALL-E changes the TTS pipeline to phoneme $\rightarrow$ discrete code $\rightarrow$ waveform, instead of phoneme $\rightarrow$ mel-spectrogram $\rightarrow$ waveform.
- ▶ The system takes as input a text prompt and a sample of the target voice, tokenizes both (using phonemes (grapheme-to-phoneme) for text and an audio codec for the speech), and uses a language model to generate discrete audio tokens for the text, matching the target speaker.
- ▶ VALL-E can do zero-shot TTS, speech editing, and content creation with other generative AI models like GPT-3.



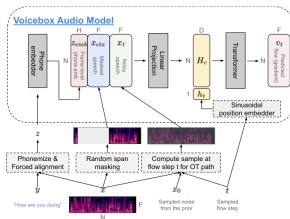
	Current Systems	VALL-E
Intermediate representation	mel spectrogram	audio codec code
Objective function	continuous signal regression	language model
Training data	≤ 600 hours	60K hours
In-context learning	X	✓

Source: Wang et al., VALL-E, Table 1: comparison with other cascaded TTS systems.

- ▶ **Data requirements:** VALL-E requires large amounts of high-quality paired text and audio data for effective training.
- ▶ **Speaker similarity:** The system relies on a reference sample; quality of voice cloning can degrade for voices and accents underrepresented in training.
- ▶ **Codec/token bottlenecks:** Using discrete audio tokens (from speech codebooks) may cause artifacts or limit audio fidelity compared to raw waveform modeling.
- ▶ **Generalization:** Performance may drop for out-of-distribution prompts or novel voice/text combinations not seen in training.
- ▶ **Ethical concerns:** Zero-shot voice synthesis can enable impersonation and raises privacy and misuse risks.

- ▶ Non-autoregressive flow-matching model trained to fill in missing speech given context audio and full transcript.
 - Style comes from audio context; content comes from text.
- ▶ One model handles zero-shot TTS, denoising, editing, style transfer, and diverse sampling via in-context learning.
- ▶ Gains over VALL-E on intelligibility and speaker similarity, up to **20×** faster generation.

- ▶ Training: learn $p(x_{\text{masked}} \mid y, x_{\text{context}})$, filling a random masked span using transcript y and unmasked audio.
- ▶ Two modules: an audio model (80-dim log-mel frames, Transformer) and a duration model.
- ▶ **Inference:** Sample random noise, which is then shaped into meaningful speech representations (mel spectrogram frames) by solving a continuous flow ordinary differential equation (ODE) conditioned on both the transcript and audio context. After obtaining the predicted mel spectrogram, a neural vocoder is used to convert these mel frames into audible waveform speech, thus completing the synthesis pipeline.

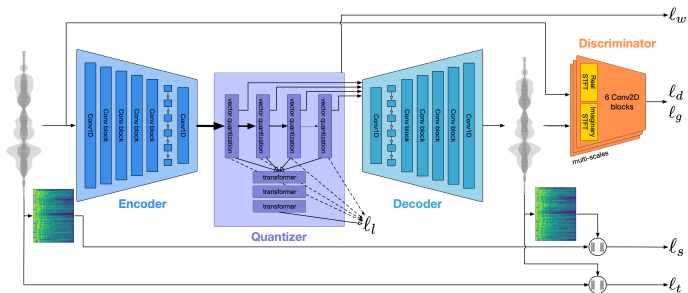


Source: Le et al., Voicebox, Fig. 1: task generalization through in-context learning; [arXiv:2306.15687](https://arxiv.org/abs/2306.15687).

- ▶ **Read speech domain:** trained mainly on audiobooks; may not transfer well to casual conversation (laughter, back-channels).
- ▶ **Front-end dependency:** needs a phonemizer and forced aligner; word-based phonemizers miss context-sensitive pronunciation.
- ▶ **Limited control:** can transfer overall style from prompts, but cannot separately pick voice from one clip and emotion from another.
- ▶ **Bias risk:** performance can drop for underrepresented accents unless training data stays diverse.
- ▶ **Misuse:** high-quality cloning raises deepfake concerns; the authors release a strong synthetic-speech classifier as a mitigation.

Model	Paradigm	Representation	Highlight
Tacotron 2	AR seq2seq	Mel	Strong naturalness
FastSpeech 2	NAR Transformer	Mel	Fast; explicit prosody
Glow-TTS	Flow + MAS	Mel	Hard alignments
Grad-TTS	Diffusion	Mel	Quality/speed trade-off knob
VITS	VAE + flow + GAN	Waveform (end-to-end)	No separate vocoder
VALL-E	Codec LM	Discrete codes	Zero-shot cloning
Voicebox	Flow matching	Continuous audio	Infilling / editing

- ▶ Neural audio codecs compress a waveform into a short sequence of **discrete tokens** and reconstruct audio from them: SoundStream (Zeghidour et al., 2021) and EnCodec (Défossez et al., 2022).
- ▶ An encoder downsamples the waveform, **residual vector quantization** maps each frame to entries of learned codebooks, and a decoder reconstructs the waveform from the codes.
- ▶ The result: audio becomes a token sequence, the same kind of object a language model already consumes and produces.

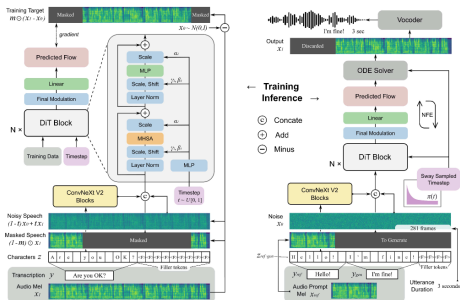


- ▶ Once audio is tokens, speech tasks become language modeling. VALL-E is the TTS instance: predict codec tokens conditioned on text and a short voice prompt.
- ▶ **AudioLM** (Borsos et al., 2022) showed pure audio continuation: a decoder-only model over audio tokens that continues speech or music with no text at all.
- ▶ **Speech-LLMs** put both directions in one model: ASR is audio tokens in, text out; TTS is text in, audio tokens out; the same transformer serves both.
- ▶ End-to-end voice conversation models take this to its conclusion: speech tokens in, speech tokens out, with no intermediate text transcript required.

- ▶ The classic voice assistant is a **cascade**: ASR $\rightarrow$ text LLM $\rightarrow$ TTS, three models and two conversions per turn.
- ▶ **Latency**: each stage waits for the previous one; a single token-level model can start responding while still listening.
- ▶ **Information loss**: a transcript discards tone, emphasis, emotion, and speaker identity; token pipelines keep them end to end.
- ▶ **Error compounding**: an ASR mistake becomes unrecoverable once only text is passed on.
- ▶ Cascades still win on modularity and controllability, which is why both designs coexist in production.

- ▶ **Flow matching won the acoustic model:** F5-TTS and CosyVoice-class systems generate speech features non-autoregressively in a few ODE steps, with zero-shot cloning from a short reference clip.
- ▶ **Codec language models power conversation:** end-to-end voice modes speak through speech tokens directly, with no separate TTS stage.
- ▶ **Open weights closed much of the gap:** XTTS, F5-TTS, and Kokoro run locally; commercial systems (ElevenLabs, OpenAI, Google) still lead on expressiveness and robustness.
- ▶ Remaining hard problems: long-form consistency, fine-grained emotion control, low-resource languages, and singing.

- Text characters (padded with filler tokens) and a masked mel-spectrogram enter diffusion-transformer blocks; the model learns a flow from noise to the missing speech, conditioned on text and the audio prompt.
- Zero-shot cloning falls out of infilling, the same idea as Voicebox, with a simpler pipeline: no phonemizer, no duration model, no forced aligner.



F5-TTS training (left) and inference (right). Chen et al., "F5-TTS: A Fairytaler that Fakes Fluent and Faithful Speech with Flow Matching," [arXiv:2410.06885](https://arxiv.org/abs/2410.06885), Figure 1.

- ▶ A voice agent chains streaming ASR, an LLM, and TTS; the user hears silence until the first audio chunk arrives, so **time to first audio** matters more than total synthesis time.
- ▶ **Streaming synthesis:** start speaking while the LLM is still writing, chunking at sentence or phrase boundaries; codec-LM stacks can stream token by token.
- ▶ **Barge-in:** users interrupt mid-sentence; the stack must stop playback, flush queued audio, and hand the conversation back to the LLM.
- ▶ Expressive control is moving from SSML markup toward natural-language style prompts (“speak warmly, a little slower”).

- ▶ **Human tests:** Does the voice sound natural?
- ▶ **Similarity:** Is the generated speech close to real speech?
- ▶ **Intelligibility:** Can speech-to-text systems understand it?
- ▶ **Cloning:** How well does it copy a specific voice?
- ▶ TTS is evaluated by playing a sentence to human listeners and having them give a mean opinion score (MOS) on a scale of 1 to 5.

- ▶ Human MOS panels are slow and expensive, so large-scale development leans on objective proxies.
- ▶ **Intelligibility:** transcribe the synthesized speech with a strong ASR model and compute WER against the input text.
- ▶ **Speaker similarity:** cosine similarity between speaker embeddings of the reference prompt and the generated audio.
- ▶ **Predicted MOS:** neural predictors (UTMOS, NISQA) estimate human scores from audio alone.
- ▶ Modern systems score close to human recordings, so plain MOS saturates; side-by-side comparative MOS (CMOS) separates systems better.

- ▶ **Consent and disclosure:** clone only with the speaker's permission and label synthetic audio as synthetic.
- ▶ **Watermarking:** embed an inaudible signature in generated audio so it can be identified later (AudioSeal).
- ▶ **Detection:** train classifiers to spot synthetic speech; Voicebox shipped one alongside the model.
- ▶ These mitigations are partial: watermarks do not survive every re-encoding, and detectors lag the newest generators.



Don't record someone without their consent

In California, all parties to any confidential conversation must give their consent to be recorded. For calls occurring over cellular or cordless phones, all parties must consent before a person can record, regardless of confidentiality.



Don't create a speech synthesizer / voice clone of someone without their consent

It might be fun, but it's a little creepy. People get upset.
Okay to use existing speech datasets (we'll provide some).



Do consider subgroup and language bias when building systems

Poor performance on subgroups
e.g. non-native speakers
Many languages are under-served relative to English/Mandarin.

Source: Stanford CS224S, Lecture 1 (2024 Spring).

- ▶ One text can be spoken many ways (speed, stress, intonation). Most systems still go text → acoustic features → waveform; newer models like VALL-E predict audio tokens directly.
- ▶ TTS improved in stages: stitch recorded clips → slow but high-quality neural models (Tacotron 2, WaveNet) → faster parallel models (FastSpeech 2, HiFi-GAN, VITS) → foundation models that clone voices from a short sample (VALL-E, Voicebox).
- ▶ Voice cloning from a few seconds of audio works well today, but can fail on rare accents; use it carefully with consent, labeling, and fake-speech detection.

1. Towards Responsible Evaluation for TTS, [arXiv:2510.06927](#).
2. Stanford CS224S: Lecture 1 (2024 Spring), [224s.24.lec1.pdf](#); (2025 Spring) course site; Jurafsky & Martin, SLP3 Ch.16.
3. Shen et al., Tacotron 2, [arXiv:1712.05884](#); van den Oord et al., WaveNet.
4. Ren et al., FastSpeech / FastSpeech 2, [1905.09263](#), [2006.04558](#).
5. Kim et al., Glow-TTS, [2005.11129](#); Popov et al., Grad-TTS, [2105.06337](#).
6. Kim, Kong, Son, VITS, [2106.06103](#); Kong et al., HiFi-GAN.
7. Wang et al., VALL-E, [2301.02111](#); Le et al., Voicebox, [2306.15687](#).
8. Zeghidour et al., SoundStream, [2107.03312](#); Défossez et al., EnCodec, [2210.13438](#); Borsos et al., AudioLM, [2209.03143](#).
9. Chen et al., F5-TTS, [2410.06885](#).
10. Saeki et al., UTMOS, [2204.02152](#); San Roman et al., AudioSeal, [2401.17264](#).
11. CMU SPSS / intro TTS materials (Black, Muthukumar); ESPnet: [github.com/espnet/espnet](#).